

# PURBANCHAL UNIVERSITY

2022

B.E. (Civil, Computer/E&C/Electrical)/Third Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

**BEG201SH: Mathematics-III** (New Course)

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

## Group A

**Answer FIVE questions.**

**5×5=25**

1. Examine the linear dependence and find the relation if dependent.  $v_1 = (1, 2, 4), v_2 = (2, -1, 3), v_3 = (0, 1, 2), v_4 = (-3, 7, 2)$
2. Prove that

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} =$$

$$2(a+b+c)^3$$

3. Solve by Gauss elimination method.

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x - y + 3z = 9.$$

4. Define Hermitian and Skew-Hermitian matrices. Show that the

$$\text{matrix } \begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix} \text{ is Hermitian.}$$

If A is Hermitian matrix, then show that I A is skew-Hermitian matrix.

5. Find the eigen values and eigen vectors of the following.

Contd. ...

(2)

Matrix.  $\begin{bmatrix} 2 & -2 & 2 \\ 1 & 1 & 1 \\ 1 & 3 & -1 \end{bmatrix}$ .

6. Find the rank of the matrix  $\begin{bmatrix} 1 & 3 & -2 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$  by reducing it

normal form.

**Group B**

**Answer TWO questions.**

**2×10=20**

- 7(a) If  $L\{f(t)\} = F(s)$ , then prove that  $L\{f'''(t)\} = s^3F(s) - s^2f(0) - sf'(0) - f''(0)$ .  
(b) Find the Laplace transform of (Any ONE)

(i)  $\frac{e^{-4t} \sin 3t}{t}$

(ii)  $\int_0^t te^t \sinh^3 t$

- 8(a) Find the inverse Laplace transform of (Any ONE)

(i)  $\frac{s^2 + 3}{s(s^2 + 9)}$

(ii)  $\frac{s^2}{(s^2 + a^2)^2}$

- (b) Use convolution theorem to evaluate

$$L^{-1} \left\{ \frac{1}{s^2(s^2 + a^2)} \right\}$$

9. The current  $i_1$  and  $i_2$  in mesh are given by the differential equation  $\frac{di_1}{dt} - wi_2 = a \cos pt$ ,  $\frac{di_2}{dt} + wi_1 = a \sin pt$ . Find the currents  $i_1$  and  $i_2$  by laplace transform if  $i_1=i_2=0$  at  $t=0$ .

**Group C**

**Answer FIVE questions.**

**5×5=25**

10. Show that the vector field

**Contd. ...**



(3)  
 $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$ . is irrotational. Also find a scalar potential  $\phi$  such that  $\vec{F} = \nabla \phi$ .

11. Find the volume of surface  $\left(\frac{x}{a}\right)^{2/3} + \left(\frac{y}{b}\right)^{2/3} + \left(\frac{z}{c}\right)^{2/3} = 1$  where  $x \times y \times z \leq 1$  by Ditchlet's theorem.

12. Find the flux of  $\vec{F}$  throughout the surface  $x^2 + y^2 + z^2 = 9$ ,  $Z \geq 0$  where  $\vec{F} = 3xi + 3yj + zk$ . ~~where  $\vec{F} = 3x1 + 3yj + zk$ .~~

13. State and prove the Green's Theorem in the plane.

14. Show that Stoke's theorem in a plane represent Green's theorem. and prove from stoke's theorem.

$$\int_C \vec{dr} \times \vec{F} = \iint_S (\hat{n} \times \nabla) \times \vec{F} ds.$$

15. Verify Gauss-Divergence Theorem for the function  $\vec{F} = yi + xj + z^2k$  over the cylindrical region bounded by  $x^2 + y^2 = 9$ ,  $z = 0$  and  $z = 2$ .

### Group D

**Answer TWO questions.**

**2×5=10**

16. Find the half-range sine series for  $e^x$  in  $0 < x < 1$

17. Find the Fourier series of the function  $f(x)$  defined

$$\text{by: } f(x) = \begin{cases} -1 & \text{for } 0 < x < \frac{\pi}{2} \\ 0 & \text{for } \frac{\pi}{2} < x < 2\pi \end{cases}$$

18. Find the Fourier series of the function  $f(x) = e^{-x}$  in  $(-1, 1)$  in complex form.



# PURBANCHAL UNIVERSITY

2014 (New)

B.E. (Computer/Electronics & Comm./Electrical)/Third Semester/Final  
Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

**BEG201SH: Mathematics-III**

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

## Group A

**Answer FIVE questions.**

**5×5=25**

1. Define Hermitian and Skew-Hermitian matrix with example and show that if  $A$  is a Hermitian then  $iA$  is a skew Hermitian matrix.

2. Prove that 
$$\begin{vmatrix} 1+a^2-b^2 & 2ab & -2b \\ 2ab & 1-a^2+b^2 & 2a \\ 2b & -2a & 1-a^2-b^2 \end{vmatrix} = (1+a^2+b^2)^3$$

3. Show that the transformation which transform  $y_1 = x_1 - x_2 + x_3$ ,  $y_2 = 3x_2 - x_2 + 2x_3$ ,  $y_3 = 2x_1 - 2x_2 + 3x_3$  is regular. Find the inverse transformation.

4. Test consistency and solve the system of equations by Gauss  
 $4y + 3z = 13$

Elimination method  $x - 2y + z = 3$ .

$$3x + 5y = 11$$

5. Find the rank of matrix 
$$\begin{bmatrix} 3 & -4 & -1 & 2 \\ 1 & 7 & 3 & 1 \\ 5 & -2 & 5 & 4 \\ 9 & -3 & 7 & 7 \end{bmatrix}$$
 by reducing to

normal form.

6. Find the Eigen values and Eigen vectors of the matrix

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}.$$

Contd. ...



**Group B****Answer FOUR questions.****4×5=20**

7. Define the Laplace transform. Show that the existence of Laplace transform if  $\lim_{s \rightarrow \infty} F(s) = 0$ .
8. Find the Laplace transform of any TWO:
- (a)  $\int_0^t e^{-t} \cos t dt$       (b)  $te^{-3t} \cos 5t$       (c)  $f(t) = \begin{cases} t-1, & \text{for } 1 < t < 2 \\ 0 & , \text{otherwise} \end{cases}$
9. Find the inverse transform of any TWO:
- (a)  $\frac{s^3}{s^4 + a^4}$       (b)  $\log\left(1 - \frac{a^2}{s^2}\right)$       (c)  $\frac{1}{(s-2)(s+2)^2}$
10. Solve the simultaneous equation:
- $$\frac{dx}{dt} + 5x - 2y = t, \quad \frac{dy}{dt} + 2x + y = 0,$$
- where,  $x(0)=y(0)=0$ , by Laplace transform method.
11. Solve the following problem by using Laplace transform method:
- $$y'' + y = \sin 3t, \quad y(0) = y'(0) = 0$$

**Group C****Answer FIVE questions.****5×5=25**

12. Find the work done in moving a particle once around a circle  $x^2 + y^2 = 4, z = 0$ , if the force field  $\vec{F}$  is defined  $\vec{F} = (2x - y + 2z)\vec{i} + (x + y - z)\vec{j} + (3x - 2y + 5z)\vec{k}$ .

**Or**

Evaluate the volume integral  $\iiint_V \vec{F} dV$  where  $V$  is the region bounded by the surface  $x=0, y=0$ .

$$y = 6, z = x^2, z = 4 \text{ and } \vec{F} = 2xz\vec{i} - x\vec{j} + y^2\vec{k}$$

13. Evaluate  $\iint_S \vec{F} \cdot \hat{n} ds$  where  $\vec{F} = 6z\vec{i} - 4\vec{j} + y\vec{k}$  and  $S$  is the surface of the plane  $2x + 3y + 6z = 12$  in the first octant.

**Contd. ...**

14. State Stoke's theorem and from it deduce the result  

$$\int_C \phi d\vec{r} = \iint_S (\vec{n} \times \nabla \phi) ds.$$
15. Find the line integral  $\int_C \vec{F} \cdot d\vec{r}$  using Stoke's theorem, where  
 $\vec{F} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$  and 'C' is the circle  $x^2 + y^2 = 1$ , which is the projection of the sphere  $x^2 + y^2 + z^2 = 1$ .
16. Find the area of the ellipse  $x^2/a^2 + y^2/b^2 = 1$  by Green's theorem.
17. Applying Gauss Divergence theorem prove that  $\iiint_S \vec{F} \cdot \hat{n} ds = \frac{3}{8}$ ,  
 where  $\vec{F} = yzi\vec{i} + zxj\vec{j} + xyk\vec{k}$  and S is the surface of sphere  $x^2 + y^2 + z^2 = 1$  in the first octant.

**Group D**

**Answer TWO questions.**

**2×5=10**

18. Define even and odd function. Find the Fourier series of the function  $f(x) = (\pi - x)$  in the interval  $[0, 2\pi]$ .
19. Obtain the half range Cosine series for  $f(x) = x$  in  $0 < x < 2$ .
20. Find the Fourier series of  $f(x) = e^{-x}$  in  $(-1 \text{ to } 1)$  in complex form.





**PURBANCHAL UNIVERSITY**

**2013 (Old)**

B.E. (Civil/Computer/Electronics & Comm.)/Third Semester/Chance

Time: 03:00 hrs.

Full Marks: 80 / Pass Marks: 32

**BEG201HS: Engineering Mathematics-III**

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

**Group A**

**Answer THREE questions.**

**3×10=30**

1(a) Prove that

$$3A^2 - 2A + 5I = \begin{bmatrix} 21 & -23 & 15 \\ -13 & 34 & 10 \\ -9 & 22 & 25 \end{bmatrix} \text{ where } A = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 1 \\ 3 & -5 & 2 \end{bmatrix}$$

(b) Prove that:

$$\begin{vmatrix} a+b & a & b \\ a & a+c & c \\ b & c & b+c \end{vmatrix} = 4abc$$

2(a) If  $A = \begin{bmatrix} 2+i & 3 & -1+3i \\ -5 & i & 4-2i \end{bmatrix}$  Show that  $A^\theta A$  is Hermitian Matrix.

(b) Express the matrix  $\begin{bmatrix} 3 & -2 & 6 \\ 2 & 7 & -1 \\ 5 & 4 & 0 \end{bmatrix}$  as the sum of a symmetric and skew symmetric matrix.

3(a) Show that the equations  $x-y+2z=5$ ,  $3x+y+z=8$  and  $2x-2y+3z=7$  are consistent and solve by Gauss elimination method.

(b) Find the inverse of  $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$  by the Cayley Hamilton theorem.

**Contd. ...**

(2)

4(a) Find the eigen values and eigen vectors of the matrix  $\begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$

(b) Find the rank of the matrix  $\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$  reducing echelon form.

### Group B

Answer TWO questions.

2×10=20

5(a) Define the Laplace Transform. If  $f(t)$  and  $f'(t)$  are continuous and  $L[f(t)] = f(s)$  then show that

$$L[f'''(t)] = S^3 F(s) - S^2 f(0) - S f'(0) - f''(0)$$

(b) Find the Laplace transform of  $f(t) = \cos\left(t - \frac{2\pi}{3}\right), t > \frac{2\pi}{3}$   
 $= 0, \quad t < \frac{2\pi}{3}$

6. Find the inverse Laplace transform of

(i)  $\frac{\sin 2t - \sin 3t}{t}$  (ii)  $\frac{\cos 2t - \cos 3t}{t}$

7. Solve the initial value problem by Laplace transform method

$$\frac{d^2 x}{dt^2} + 2 \frac{dx}{dt} + 2x = e^{-t} \sin t, x(0) = 0, x'(0) = 1$$

### Group C

Answer THREE questions.

3×10=30

8(a) Evaluate  $\int_C \vec{F} \cdot d\vec{r}$  where  $\vec{F} = (x - y)\vec{i} + (x + y)\vec{j}$  along the curve bounded by  $y = \sqrt{x}$  and  $y = x^2$

(b) Evaluate  $\int_S \vec{F} \cdot \hat{n} \, ds$  where  $\vec{F} = (x + y^2)\vec{i} - 2x\vec{j} + 2yz\vec{k}$  and  $S$  is the surface of the plane  $2x + y + 2z = 6$  in the first octant.

Contd. ...

(3)

9(a) Find the volume bounded by the surface  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^4}{c^4} = 1$  by Dirichlet's theorem.

(b) Evaluate  $\iiint_V \vec{F} dv$  where  $v$  is the volume of region bounded by  $x=0, y=0, y=6, z=x^2$  and  $z=4$  and  $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$

10(a) State and prove Green's theorem in the plane.

(b) Prove  $\oint_C \vec{dr} \times \vec{F} = \iiint_S (\hat{n} \times \nabla) \times \vec{F} ds$  by using Stoke's theorem.

11(a) Verify the Stoke's theorem for the function  $\vec{F} = y\vec{i} + z\vec{j}$  over the plane surface  $2x + 2y + z = 1$  lying in the 1<sup>st</sup> octant.

(b) Evaluate by Gauss Divergence theorem for  $\iiint_S \vec{F} \cdot \hat{n} ds$ , where

$\vec{F} = x\vec{i} - y\vec{j} + (z^2 - 1)\vec{k}$  and  $S$  is the cylinder formed by the surface  $z=0, z=1$  and  $x^2 + y^2 = 4$



# PURBANCHAL UNIVERSITY

2013

B.E. (Civil/Computer/Electronics & Comm.)/Third Semester/Back

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

**BEG201HS: Engineering Mathematics-III (Old Course)**

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

## Group A

Answer **THREE** questions.

3×10=30

1(a) Prove that 
$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc \left( 1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$$

(b) Show that  $(AB)^{-1} = B^{-1}A^{-1}$  where A and B are two non-singular matrices of same order.

2(a) If  $A = \begin{vmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 1 & 0 & 1 \end{vmatrix}$ . Find the value of  $A^2 - 5A + 2I$ , where I is 3×3

unit matrix.

(b) Define Hermitian and skew-hermitian matrix. Prove that the

matrix  $\begin{vmatrix} 1 & 1-i & 2 \\ 1+i & 3 & i \\ 2 & -i & 0 \end{vmatrix}$  is Hermitian-matrix.

3(a) Find the rank of the matrix by reducing normal form:

$$\begin{bmatrix} 1 & 3 & -2 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$$

Contd. ...

(2)

- (b) Solve the following system of linear equations by Gauss Elimination method:

$$2x - 2y + z = 1$$

$$x + 2y + 2z = 2$$

$$2x + y - 2z = 7$$

- 4(a) Find the eigen values and eigen vectors of the matrix  $\begin{bmatrix} 3 & -1 \\ 1 & 1 \end{bmatrix}$ .

- (b) Verify Cayley-Hamilton theorem for the matrix and find its

inverse  $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ .

**Group B**

**Answer TWO questions.**

**2×10=20**

- 5(a) Find the laplace transform of (any TWO):

(i)  $te^{2t} \sin t$

(ii)  $\frac{1 - \cos t}{t}$

(iii)  $\sin at \cos at$ .

- (b) State and prove linear property of laplace transform.

6. Find the inverse laplace transform of (any TWO):

(i)  $\frac{s+4}{s(s-1)(s^2+4)}$

(ii)  $\log\left(\frac{s^2+1}{s^2}\right)$

(iii)  $\frac{s^2}{(s^2+a^2)^2}$

7. Solve the differential equation

$y'' + 9y = \cos 2t$ ;  $y(0) = 1$ ,  $y'(0) = -1$  using laplace transformation method.

**Contd. ...**



**(3)**  
**Group C**

**Answer THREE questions.**

**3×10=30**

- 8(a) Prove that the line integral  $\int_C \vec{F} \cdot d\vec{r}$  is independent of the path joining any two points in the region if and only if  $\vec{F} = \nabla \phi$  where  $\phi$  is a scalar point function.
- (b) Evaluate  $\int_C \vec{F} \cdot d\vec{r}$  where  $\vec{F} = (x^2 + y^2)\vec{i} + (x^2 - y^2)\vec{j}$  and C is the curve  $y = x^3$  from (0, 0) to (1, 1).
- 9(a) Evaluate  $\iint_S \vec{F} \cdot \hat{n} ds$ , where  $\vec{F} = z\vec{i} + x\vec{j} - yz\vec{k}$ , and S is the surface of the cylinder  $x^2 + y^2 = 9$  included in the first octant between  $z = 0$  and  $z = 4$ .
- (b) Find the volume of the surface  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$  by Dirichlet's theorem.
- 10(a) State and prove Green's theorem in plane.
- (b) Use Green's theorem to find the area of the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ .
- 11(a) Evaluate  $\iint_S \vec{F} \cdot \hat{n} ds$  by using Gauss divergence theorem for  $\vec{F} = xy\vec{i} + z^2\vec{j} + 2yz\vec{k}$  and S is the surface bounded by  $x = 0$ ,  $y = 0$ ,  $z = 0$  and the plane  $x + y + z = 1$ .
- (b) Verify stoke's theorem for a vector field  $\vec{F} = x^2\vec{i} + xy\vec{j}$  taken round the rectangle in the xy - plane bounded by  $x = 0$ ,  $x = a$ ,  $y = 0$ ,  $y = b$ .





**PURBANCHAL UNIVERSITY**

**2012**

B.E. (Civil/Computer/Electronics & Comm.)/Third Semester/Back

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

**BEG201HS: Engineering Mathematics-III**

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

**Group A**

**Answer THREE questions.**

**3×10=30**

1(a) If  $A = \begin{bmatrix} 1 & 3 & 2 \\ 0 & 5 & -1 \\ 2 & 3 & 0 \end{bmatrix}$ , evaluate  $A^2 - 2A + 5I$ , where I is unit matrix of order 3.

(b) Prove that:

$$\begin{vmatrix} (a+b)^2 & ca & bc \\ ca & (b+c)^2 & ab \\ bc & ab & (c+a)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

2(a) Find whether the vectors  $x_1 = (3,2,7), x_2 = (2,4,1), x_3 = (1,-2,6)$  are linearly dependent or independent.

(b) Solve by Gauss-Elimination method:

$$x - 6y - z = 10$$

$$2x - 2y + 3z = 10$$

$$3x - 8y + 2z = 20$$

3(a) Show that the transformation

$y_1 = 2x_1 + x_2 + x_3, y_2 = x_1 + x_2 + 2x_3, y_3 = x_1 - 2x_3$  is regular, write down the inverse transformation.

(b) Find the Eigen values and eigen vectors of the matrix.

**Contd. ...**

(2)

$$\begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

4. Verify Cayley- Hamilton theorem for the following matrix and find its inverse.

$$\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

**Group B**

**Answer TWO questions.**

**2×10=20**

- 5(a) State and prove first shifting property of Laplace transform.

- (b) Find the Laplace transform of

(i)  $te^{-t} \sin t$

(ii)  $\frac{1-e^{-t}}{t}$

- 6(a) Find the inverse Laplace transform of  $\frac{s}{(s-3)(s^2+4)}$

- (b) Apply the convolution theorem to evaluate  $L^{-1}\left\{\frac{s}{(s^2+1)(s^2+4)}\right\}$

7. Solve the simultaneous equations:

$$\frac{dx}{dt} + 5x - 2y = t, \quad \frac{dy}{dt} + 2x + y = 0 \quad \text{being given } x = y = 0 \text{ when } t = 0$$

by the transform method.

**Group C**

**Answer THREE questions.**

**3×10=30**

- 8(a) Find the work done in moving particle in the field

$$\vec{F} = (3x - 4y + 2z)\vec{i} + (4x + 2y - 3z^2)\vec{j} + (2xz - 4y^2 + z^2)\vec{k} \quad \text{along}$$

one round of the ellipse  $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1, z = 0$ .

**Contd. ...**

(3)

- (b) Prove that  $\vec{F} = (2xz^3 + 6y)\vec{i} + (6x - 2yz)\vec{j} + (3x^2z^2 - y^2)\vec{k}$  is conservative vector field. Also find scalar potential function.
- 9(a) Evaluate  $\iint_S \vec{F} \cdot \hat{n} \, ds$  where  $\vec{F} = x\vec{i} + y\vec{j} + z\vec{k}$  and S is the surface of the sphere  $x^2 + y^2 + z^2 = 9$ .
- (b) Evaluate  $\iiint x^{l-1} y^{m-1} z^{n-1} dx dy dz$  where x, y, z are all positive but  $\left(\frac{x}{a}\right)^p + \left(\frac{y}{b}\right)^q + \left(\frac{z}{c}\right)^r \leq 1$ .
- 10(a) State the Stoke's Theorem and hence derive the Green's Theorem in a plane as a special case of Stoke's Theorem.
- (b) Apply Stoke's theorem to evaluate  $\int_C (x+y)dx + (2x-z)dy + (y+z)dz$  where C is the boundary of the triangle with vertices  $(2,0,0)$ ,  $(0,3,0)$  and  $(0,0,6)$ .
- 11(a) Prove that  $\iint_S (\hat{n} \cdot \vec{F}) \, ds = \iiint_V (\nabla \cdot \vec{F}) \, dV$
- (b) Verify Gauss divergence theorem for the function  $\vec{F} = y\vec{i} + x\vec{j} + z^2\vec{k}$  over the cylindrical region bounded by  $x^2 + y^2 = 9$ ,  $z = 0$  and  $z = 2$ .

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PORBANCHAL UNIVERSITY

2012

B.E. (Civil/Computer/Electronics & Comm.)/Third Semester/Final

Time: 03:00 hrs.

Full Marks: 80 / Pass Marks: 32

BEG103HS: Engineering Mathematics-III

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Group A

Answer ALL questions.

16×5=80

1 Prove that

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

2 Define Hermitian and Skew-Hermitian matrices show that the matrix,

$$\begin{bmatrix} 3 & 7-4i & -2+5i \\ 7+4i & -2 & 3+i \\ -2-5i & 3-i & 4 \end{bmatrix} \text{ is Hermitian.}$$

3 Solve the following equations

$$x - 2y + 3z = 2$$

$$2x - 3y + z = 1$$

$$3x - y + 2z = 9$$

by Gauss elimination method.

4 Find the rank of the matrix,

$$\begin{bmatrix} 2 & 4 & -4 & 3 \\ 1 & -2 & -1 & 1 \\ 1 & 2 & -1 & 3 \end{bmatrix}$$

5 For any two non-singular matrices A and B, prove that  $(AB)^{-1} = B^{-1}A^{-1}$

Contd. ...



(2)

Or,

Find the eigen values and eigen vectors of the matrix,

$$\begin{bmatrix} 1 & 0 & -1 \\ 1 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix}$$

6 Find the Laplace transform of (any TWO)

(i)  $t \sinh t$

(ii)  $\frac{1-e^t}{t}$

(iii)  $\cos(\omega t + \theta)$

7. Find the inverse laplace transform for (any TWO)

(i)  $\frac{1}{S^2 - 5S + 6}$

(ii)  $\log \frac{S+1}{S-1}$

(iii)  $\tan^{-1} \frac{2}{S}$

8 Solve the following differential equation by the method of transforms.

$$y'' - 4y = 0, y(0) = 1, y'(0) = 0$$

OR

$$y'' + y = \sin 3t, y(0) = y'(0) = 0$$

9 If  $L\{f(t)\} = F(s)$ , Prove that

$$L\{f''(t)\} = S^2 F(s) - S f(0) - f'(0)$$

10 Prove that the line integral  $\int_A^B \vec{F} \cdot d\vec{r}$  is independent of the path joining any two points A and B in a region if and only if  $\oint_C \vec{F} \cdot d\vec{r} = 0$  for any simple closed curve C in the region.

OR

Find the work done in moving particle once around a circle

$x^2 + y^2 = 4, z = 0$  if the force field  $\vec{F}$  is defined by

$$\vec{F} = (2x - y + 2z)\vec{i} + (x + y - z)\vec{j} + (3x - 2y - 5z)\vec{k}$$

11 If R is the region bounded by  $x \geq 0, y \geq 0$  and  $x + y \leq h$ , prove that

Contd. ...

(3)

$$\iint_R x^{l-1} y^{m-1} dx dy = \frac{\overline{l} \overline{m}}{(l+m+1)} h^{l+m}$$

OR

Evaluate  $\iint_S \vec{F} \cdot \hat{n} ds$  where

$\vec{F} = z\vec{i} + x\vec{j} - yz\vec{k}$  and S is the surface of the cylinder  $x^2 + y^2 = 9$  included in the first octant between  $z = 0$  and  $z = 4$ .

12. Use green's theorem, find the area of the asteroid  $x^{2/3} + y^{2/3} = a^{2/3}$

13. Applying Gauss Divergence theorem prove that  $\iiint_S \vec{r} \cdot d\vec{s} = 3v$

where v is the volume enclosed by a closed surfaces.

14. Verify stroke's theorem for

$\vec{F} = (2x - y)\vec{i} - yz^2\vec{j} - y^2z\vec{k}$  where S is the surface of the sphere  $x^2 + y^2 + z^2 = 1$  above the xy plane and C is its boundary

15. Find the fourier series of the function

$$f(x) = \begin{cases} 0 & \text{for } x \leq x < \frac{\pi}{2} \\ 1 & \text{for } \frac{\pi}{2} \leq x < \pi \\ 0 & \text{for } \pi \leq x \leq 2\pi \end{cases}$$

16. Define even and odd function. Find the fourier series of the function  $f(x) = |x|$  in the interval  $(-2, 2)$

OR

Find the half range fourier sine series of the function

$f(x) = e^x$  in the interval  $(-1, 1)$

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# PURBANCHAL UNIVERSITY

2011

B.E. (Civil/Computer/Electronics & Comm.)/Third Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

**BEG201HS: Engineering Mathematics-III**

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

## Group A

**Answer THREE questions.**

**3×10=30**

- 1(a) Define orthogonality of a matrix. For any two orthogonal matrices A and B, show that AB is an orthogonal matrix. What is the nature of BA?

(b) Evaluate 
$$\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

- 2(a) If  $A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$ . Find the value of  $A^2 - 5A + 2I$ , where I is 3×3 unit matrix.

- (b) Define Hermitian and skew Hermitian matrix and give example. If A is a Hermitian matrix then show that iA is a skew-Hermitian matrix.

- 3(a) Find the eigen values and eigen vectors of the matrix  $A = \begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$

Also, find the eigen values of  $A^{-1}$  and  $A^2$ .

- (b) Find the characteristic equation and use cayley-Hamilton

theorem to find the inverse of the matrix  $\begin{bmatrix} 7 & -1 & 3 \\ 6 & -1 & 4 \\ 2 & 4 & 8 \end{bmatrix}$ .

**Contd. ...**

- 4(a) Solve the following system of linear equations by Gauss elimination method:

$$x + y + z = 8$$

$$x - y + 2z = 6$$

$$3x + 5y - 7z = 14$$

- (b) Find the rank of the following matrix, reducing it to the normal form:

$$\begin{bmatrix} 1 & 3 & -2 & 1 \\ 1 & 1 & 1 & 2 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$$

### Group B

Answer TWO questions.

2×10=20

- 5(a) Define unit step function. Find the Laplace transform of  $f(t)$  defined as

$$f(t) = \cos\left(t - \frac{2\pi}{3}\right); \quad t > \frac{2\pi}{3}$$

$$= 0, \quad t < \frac{2\pi}{3}$$

- (b) Find the Laplace transform of

(i)  $\frac{1 - \cos t}{t}$

(ii)  $\cosh at \cos at$

- 6(a) Find the inverse Laplace transform of

(i)  $\frac{3s + 5\sqrt{2}}{s^2 + 8}$

(ii)  $\tan^{-1} \frac{2}{s}$

- (b) State convolution theorem and use it to obtain the inverse Laplace

transform of  $\frac{s^2}{(s^2 + a^2)(s^2 + b^2)}$

7. Solve the equation, using Laplace transform method:

$$\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = te^{-t}, \quad y(0) = 1, \quad y'(0) = 0$$

Contd. ...



**Answer THREE questions.****3×10=30**

8(a) Prove that the line integral  $\int_C \vec{F} \cdot d\vec{r}$  is independent of the path

joining any two points in the region if and only if  $\vec{F} = \nabla \phi$  where  $\phi$  is a scalar function.

(b) Find the work done in moving a particle once round the circle  $x^2 + y^2 = 9, z = 0$  under the force field given by

$$\vec{F} = (2x - y + z)\vec{i} + (x + y - z)\vec{j} + (3x - 2y + 4z)\vec{k}$$

9(a) Evaluate  $\int_S \vec{F} \cdot \hat{n} \, ds$ , for  $\vec{F} = (x^2 + y^2)\vec{i} - 2xy\vec{j}$  where S is the surface bounded by the cylinder  $x^2 + y^2 = 4$  between planes  $z = 0$  and  $z = 1$ .

(b) Find the volume of the surface  $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^4}{c^4} \leq 1$  by using the Dirichlet's theorem.

10(a) By using Green's theorem prove that the area of the plane curve is  $\frac{1}{2} \int_C (x dy - y dx)$  and hence find the area of the hypocycloid

$$\frac{x^{2/3}}{a^{2/3}} + \frac{y^{2/3}}{b^{2/3}} = 1$$

(b) Verify Stoke's theorem for a vector field  $\vec{F} = x^2\vec{i} + xy\vec{j}$  taken round the rectangle in the xy-plane bounded by  $x = 0, x = a, y = 0, y = b$

11(a) Prove that: (i)  $\iint_S (\hat{n} \times \vec{F}) \, ds = \iiint_V (\nabla \times \vec{F}) \, dv$  (ii)  $\iiint_V \nabla^2 r^2 \cdot \hat{n} \, ds = 6V$

(b) Evaluate using Gauss-Divergence theorem:

$$\iiint_S [(x^3 - yz)\vec{i} - 2x^2y\vec{j} + 2z\vec{k}] \cdot \hat{n} \, ds, \text{ where S is the surface of the}$$

cube bounded by the planes  $x = 0, x = a, y = 0, y = a, z = 0, z = a$ .

Candidates are required to give their answers in their own words as far as practicable.

All questions carry equal marks. The marks allotted for each sub-question is specified along its side.

Answer FIVE questions.

- 1(a) How D operator can be used to describe the given circuit in terms of differential equation? Obtain the differential equation that relates  $e$  to  $i$ . 8

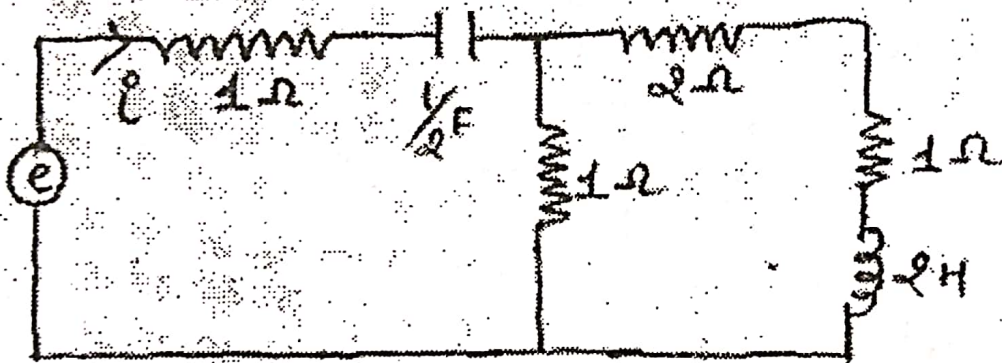


Fig. 1(a)

- (b) Draw the waveform of step voltage response for the first order RC circuit with the explanation of significance of time constant in such circuits while the input being square wave. For the below figure find the complete solution for  $i$ . 8

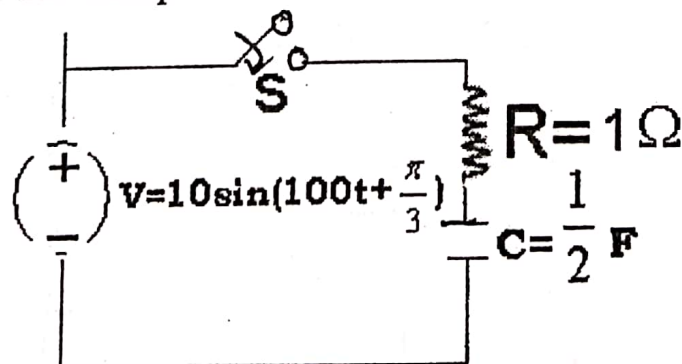


Fig. 1(b)

Contd. ...

(2)

- 2(a) State and prove initial and final value theorem. Find the initial value of current given 5

$$I(S) = (2S + 5) / \{(S + 1)(S + 2)\}$$

- (b) Solve the following differential equation using laplace transformation method: 5

$$d^2i/dt^2 + di/dt = t^2 + t. \text{ It is given that } i(0^+) = 4A, di(0^+)/dt = -2\text{Amp/S.}$$

- (c) An exponential current of  $4e^{-t}$  Ampere is applied at time  $t=0$  to a parallel R-L circuit comprising  $R=1\Omega$  and  $L=0.33H$ . Using laplace transformation method, obtain the complete solution for  $V(t)$  across the parallel circuit. Assume zero current in the inductor  $L$  before the application of current. 6

- 3(a) Mention the different possible transfer ratios for relating voltages and currents for the given two port network. 5

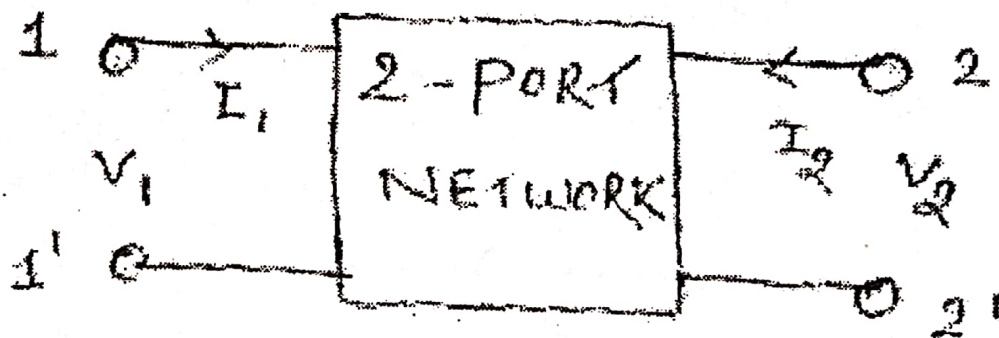


Fig. 3(a)

- (b) For the given two port network determine the voltage transfer ratio  $G_{21}(S)$ . If this network is terminated at port 2 with a  $2\Omega$  resistor, find  $Z_{11}(S)$  and  $Z_{21}(S)$ . 6

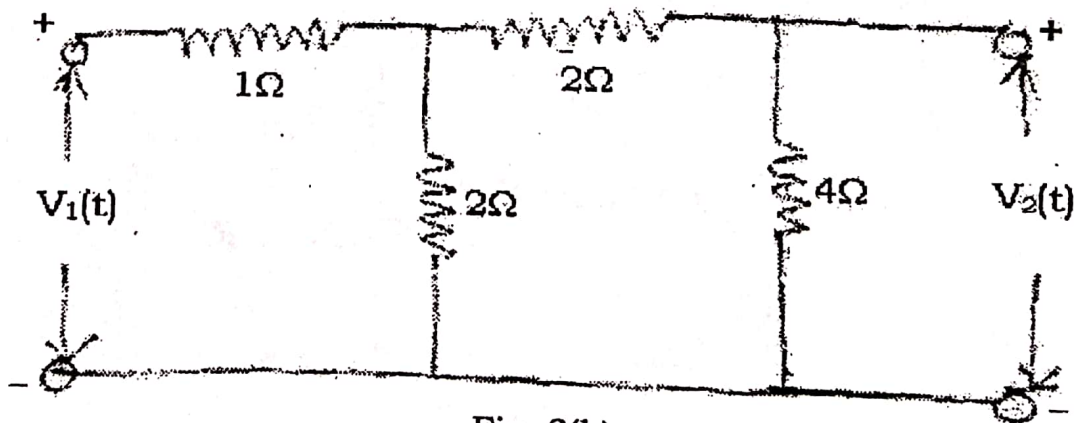


Fig. 3(b)

Contd. ...



(3)

- (c) What is Routh-Hurwitz theorem and its criterion? Check the stability of the given network function using R-H test: 5

$$F(S) = (S+2)(S^2+9)/(S^2+4)(S^2+4S+5)$$

- 4(a) Find the Fourier Series of the below waveform: 8

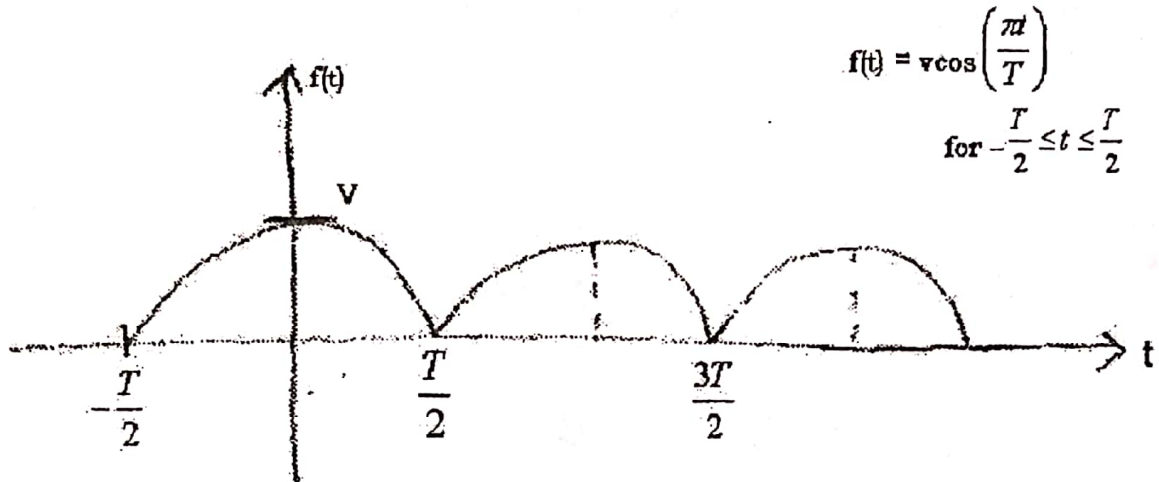


Fig. 4(a)

- (b) Using R-H criterion, determine the range of k for stability. 8

$$s^4 + 2s^3 + (4+k)s^2 + 9s + 50 = 0$$

- 5(a) What do you mean by lossless network? What are its properties? 9

For the given network function:

$$Z(S) = 10 \frac{(S^2+4)(S^2+16)}{(S)(S^2+9)}$$

- (i) State whether or not the given function LC network? Why?  
(ii) Synthesize the given network.

- (b) Find the Y-Parameter for the network shown in figure below: 7

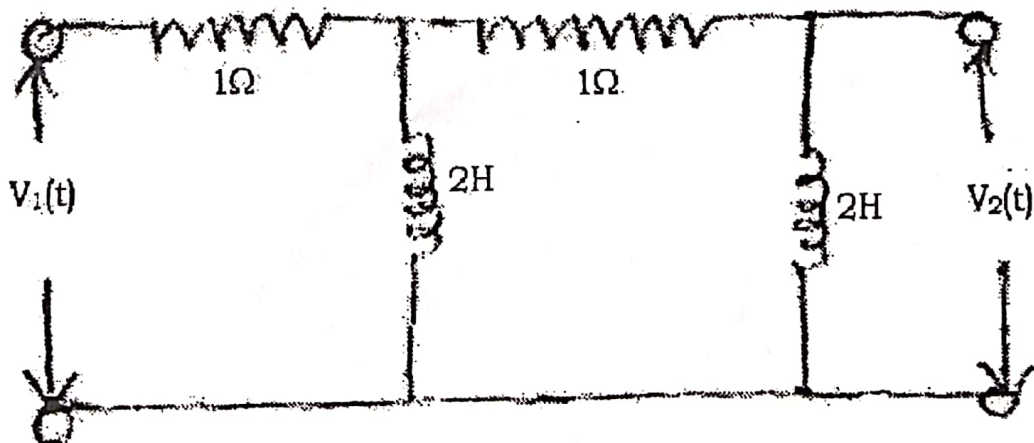


Fig. 5(b)

Contd. ...



(4)

6(a) Use Causer II method to realize the given network function: 8

$$Z(S) = (2S^4 + 20S^2 + 18)/(S^3 + 4S)$$

(b) From the Pole-Zero plot of the below figure synthesize the network using Foster I Method. Use gain factor-2 while obtaining the transfer impedance  $Z(S)$ . 8

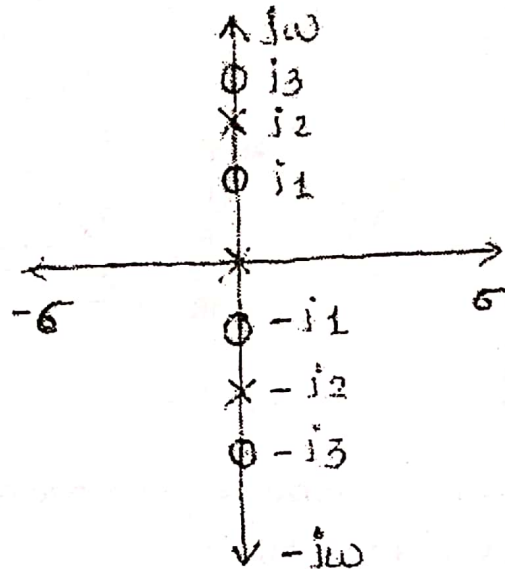


Fig. 6(b)

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**PURBANCHAL UNIVERSITY**

**2009**

B.E. (Civil/Computer/Electronics & Comm./Third Semester/Final

Time: 03:00 hrs.

Full Marks: 80 /Pass Marks: 32

**BEG201HS, Engineering Mathematics-III**

*Candidates are required to give their answers in their own words as far as practicable.*

*The figures in the margin indicate full marks.*

**Group A**

**Answer THREE questions.**

**3×10=30**

1(a) Show that 
$$\begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix} = 2abc(a+b+c)^3$$

(b) If A is any square matrix, prove that  $A+A^*$  is Hermitian and  $A-A^*$  is Skew-Hermitian.

2(a) Express the matrix  $\begin{bmatrix} 3 & -2 & 6 \\ 2 & 7 & -1 \\ 5 & 4 & 0 \end{bmatrix}$  as the sum of a symmetric and a Skews-symmetric matrix.

(b) Solve the system of equations by Gauss elimination method.

$$2x - 3y + 2z = 5$$

$$x + 2y - 3z = 4$$

$$3x + y - 4z = 7$$

3(a) Find the rank of the matrix  $\begin{bmatrix} 6 & 1 & 3 & 8 \\ 4 & 2 & 6 & -1 \\ 10 & 3 & 9 & 7 \\ 16 & 4 & 12 & 15 \end{bmatrix}$ .

**Contd. ...**

(2)

(b) Verify Cayley-Hamilton theorem for the matrix and find its

$$\text{inverse} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

4(a) Find the Eigen values and Eigen vectors of the matrix

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

(b) Show that the transformation  $y_1 = 2x_1 + x_2 + x_3$ ,  $y_2 = x_1 + x_2 + 2x_3$ ,  $y_3 = x_1 - 2x_3$  is regular.

### Group B

Answer TWO questions.

2×10=20

5(a) If  $f(t)$  and  $f'(t)$  are both continuous for  $t \geq 0$ , then show that  $\{[f'(t)]\} = s^2 F(s) - sf(0) - f'(0)$

(b) Find the Laplace transform of

(i)  $t^2 \cos at$

(ii)  $\cosh at \sin at$

6. Find the inverse Laplace transform of the function.

(a)  $\frac{s^2 + 3}{s(s^2 + 9)}$  (b)  $\frac{1}{2} \log \frac{s^2 + b^2}{s^2 + a^2}$

7. Solve the differential equation

$$\frac{d^2 x}{dt^2} + 9x = \cos 2t, x(0) = 1, x'(0) = -1 \text{ Using Laplace transform method.}$$

### Group C

Answer THREE questions.

3×10=30

8(a) Evaluate  $\int_c \vec{F} \cdot d\vec{r}$  where  $\vec{F} = z\vec{i} + x\vec{j} + y\vec{k}$  and  $c$  is the arc of the curve  $\vec{r} = \cos \theta \vec{i} + \sin \theta \vec{j} + \theta \vec{k}$  from  $\theta = 0$  to  $\theta = 2\pi$ .

Contd. ...

(3)

- (b) Show that  $\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$  is irrotational and find its scalar potential function.

- 9(a) Evaluate  $\iint_s \vec{F} \cdot \hat{n} ds$  where  $\vec{F} = 6z\vec{i} - 4\vec{i} + y\vec{k}$  and  $s$  is the surface of the plane  $2x + 3y + 6z = 12$  in the first octant.

- (b) Find the whole volume bounded by the surface

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \text{ by Dirichlet's theorem.}$$

- 10(a) State and prove Green's theorem in plane.

- (b) Using the Green's Theorem find the area of the ellipse

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

- 11(a) Verify Stoke's theorem for the function  $\vec{F} = y\vec{i} - z\vec{j}$  over the plane surface  $2x + 2y + z = 2$  lying in the 1<sup>st</sup> octant.

- (b) Evaluate  $\iint_s \vec{F} \cdot \hat{n} ds$  where  $\vec{F} = 2x\vec{i} - 3y\vec{j} + 4z\vec{k}$  and  $s$  is the surface of sphere  $x^2 + y^2 + z^2 = 1$  by Gauss's divergence theorem.

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# PURBANCHAL UNIVERSITY

## III SEMESTER FINAL EXAMINATION-2007

**LEVEL:** B.E.(Civil/Computer/Electronics & Communication/Geomatic)

**SUBJECT:** BEG201HS, Engineering Mathematics-III

**Full Marks:** 80

**TIME:** 03:00 hrs.

**Pass Marks:** 32

*Candidates are required to give their answers in their own words as far as practicable.*

*Figures in the margin indicate full marks.*

### GROUP-A

**Attempt any THREE questions.**

**[3×10=30]**

**Q. [1] [a]** Define inverse of a square Matrix. Show that the inverse of the Matrix is unique, if exists. Also for any Matrix A, prove that  $(A^T)^{-1} = (A^{-1})^T$ .

**[b]** Prove that

$$\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$$

**Q. [2] [a]** If  $A = \begin{bmatrix} -1 & 2+i & 5-3i \\ 2-i & 7 & 5i \\ 5+3i & -5i & 2 \end{bmatrix}$ ; Show that A is Hermitian

Matrix and  $iA$  is Skew-Hermitian Matrix.

**[b]** Solve by Gauss' Elimination method

$$3x - y + 2z = 1$$

$$x - y + z = -1$$

$$2x - 2y + 3z = 2$$

**Contd. ...**

(2)

Q. [3] [a] Write the Rank of the Diagonal Matrix

$$\begin{bmatrix} 8 & & & \\ & 0 & & \\ & & -7 & \\ & & & 0 \\ & & & & 8 \end{bmatrix}$$

Also Find the Rank of the following Matrix, Reducing it to the Normal form:

$$\begin{bmatrix} 1 & -1 & 2 & -3 \\ 4 & 1 & 0 & 2 \\ 0 & 3 & 1 & 4 \\ 0 & 1 & 0 & 2 \end{bmatrix}$$

[b] When the Set of  $n$  vectors ( $x_1, x_2, x_3, \dots, x_n$ ) is said to be linearly dependent? Show that the Transformation given by  $y_1 = x_1 - x_2 + x_3$ ,  $y_2 = 3x_1 - x_2 + 2x_3$ ,  $y_3 = 2x_1 - 2x_2 + 3x_3$  is non-singular. Also find the inverse transformation.

Q. [4] [a] Verify the Cayley-Hamilton Theorem for the Matrix

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

[b] Define the Characteristic equation of a square Matrix. Also Find the Eigen Values and Eigen Vectors of the Matrix

$$A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$$

Contd. ...

(3)

**GROUP-B****Attempt any TWO questions.****[2×10=20]**

Q. [5] [a] State the Linearity property of Laplace transform and hence show that  $L(e^{at} f(t)) = F(s-a)$  where  $F(s)$  is the Laplace Transform of  $f(t)$ .

[b] Find the inverse Laplace Transform of  $\frac{1 - \cos t}{t^2}$

Q. [6] [a] Find the inverse Laplace Transform of  $\frac{4s+15}{16s^2 - 25}$

[b] Apply the Convolution Theorem to Evaluate

$$L^{-1} \left\{ \frac{s}{(s^2 + a^2)^2} \right\}$$

Q. [7] Solve the following Simultaneous Equations

$$\frac{dx}{dt} - y = e^t, \quad \frac{dy}{dt} + x = \sin t \quad \text{given } x(0) = 1, y(0) = 0 \text{ by}$$

Applying the 'Convolution Technique'.

**GROUP-C****Attempt any THREE questions.****[3×10=30]**

Q. [8] [a] Show that  $\vec{F} = 2xyz^2 \vec{i} + (x^2z^2 + z \cos yz) \vec{j} + (2x^2yz + y \cos yz) \vec{k}$  is Conservative. Also find the Scalar point

function  $\phi$  such that  $\vec{F} = \nabla \phi$ . Finally evaluate  $\oint_C \vec{F} \cdot d\vec{r}$

from  $(1, 0, 1)$  to  $(0, \frac{\pi}{2}, 1)$

[b] If  $\phi = 2xyz^2$  and  $C$  be the curve  $x = t^2, y = 2t, z = t^3$

from  $t = 0$  to  $t = 1$ . Evaluate  $\oint_C \phi d\vec{r}$ .

Contd. ...

Q. [9] [a] Find the Flux of  $\vec{F}$  through the surface of sphere  $x^2 + y^2 + z^2 = 1$  above the  $xy$ -Plane where  $\vec{F} = y^2z^2 \vec{i} + z^2x^2 \vec{j} + x^2y^2 \vec{k}$ .

[b] State and prove the Ditchlet's Integral.

Q. [10] [a] State the Stoke's Theorem and Hence Derive the Green's Theorem in a plane as a special case of Stoke's Theorem.

[b] Evaluate by Stoke's Theorem  $\oint_C (e^x dx + 2y dy - dz)$  where

$C$  is the curve given by  $x^2 + y^2 = 4, z = 2$

Q. [11] [a] Prove that  $\iint_S (\vec{n} \times \vec{F}) ds = \iiint_V (\nabla \times \vec{F}) dv$

[b] As an application of Gauss Theorem show that the Flux of the radius vector  $\vec{r}$ , through any closed surface  $S$  is three times the volume bounded by that surface. (i.e Flux of  $\vec{r} = 3v$ ).

∴ Further for the same close surface  $S$ , prove

$$\iint_S (\nabla \times \vec{F}) \cdot \vec{n} ds = 0.$$





Belongs to Panukg

## PURBANCHAL UNIVERSITY

### III SEMESTER FINAL EXAMINATION-2003

LEVEL: B. E. (Civil/Computer/Electronics & Communication)

SUBJECT: BEG201HS, Mathematics-III

TIME: 03:00 hrs.

Full Marks: 80

Pass Marks: 32

Candidates are required to give their answers in their own words as far as practicable.

Figures in the margin indicate full marks.

#### GROUP-A

Attempt any THREE questions.

[3×10=30]

1. [1] [a] Show that

$$\begin{vmatrix} a+b & a & b \\ a & a+c & c \\ b & c & b+c \end{vmatrix} = 4abc$$

[b] Prove that

$(AB)^{-1} = B^{-1}A^{-1}$ , where A and B are two non-singular matrices.

[2] [a] If  $A = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

Find the value of  $A^2 - 5A + 2I$ ; where I is unit matrix.

[b] Obtain the matrix transformation which transforms the point (1, 1) into the point (-2, 0) and the point (0, 1) into (-1, 1).

Contd. ...

Q. [3] [a] Solve the equation by using Gauss-Elimination Method.

$$x + 2y - z = 1$$

$$x + y + 2z = 9$$

$$2x + y - z = 2$$

[b] Find the rank of

$$\begin{bmatrix} 1 & 3 & -2 & 1 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & -3 & 2 \\ 3 & 3 & -3 & 3 \end{bmatrix}$$

Q. [4] [a] Find the inverse matrix of

$$\begin{bmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

[b] Find the eigen values and corresponding eigen vectors of the matrix.

$$\begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 3 \end{bmatrix}$$

### GROUP-B

Attempt any TWO questions.

[2×10=20]

Q. [5] [a] State and prove existence property of Laplace Transform.

[b] Find the Laplace Transform of

[i]  $\frac{1 - \cos t}{t}$

[ii]  $e^{at+b}$



Q. [6] Find the Inverse Laplace Transform of

[a]  $\frac{2s-5}{4s^2+25} + \frac{5}{s+3}$

[b]  $\frac{1}{s^3 - a^3}$

Q. [7] Solve the differential equation by Transform method:

$$\frac{d^2 y}{dt^2} + \frac{dy}{dt} - 2y = t; y(0) = 1, y'(0) = 0$$

**GROUP-C**

Attempt any THREE questions.

[3×10=30]

Q. [8] [a] If  $\vec{F}$  is the gradient of some scalar point function  $\Phi$  that is  $\vec{F} = \nabla\Phi$ ; then the line integral  $\int_C \vec{F} \cdot d\vec{r}$  is independent of the path joining any two points in the region and conversely.

[b] Find the total work done in moving particle in a force field given by  $\vec{F} = 2xy\vec{i} - 3x\vec{j} - 5z\vec{k}$  along the curve  $x=t, y=t^2+1, z=2t^2$  as  $t$  varies from 1 to 2.

Q. [9] [a] Evaluate  $\iint_S \vec{F} \cdot \vec{n} \, ds$ , where

$\vec{F} = yz\vec{i} + zx\vec{j} + zy\vec{k}$  through the portion of sphere  $x^2+y^2+z^2=1$  in the first octant.

[b] If  $R$  is a region bounded by  $x \geq 0; y \geq 0$  and  $x+y \leq h$ ; then prove that

$$\iint_R x^{m-1} y^{n-1} \, dx dy = \frac{\int_0^m \int_0^n}{\int_0^{m+n+1}} h^{m+n}.$$

Contd. ...

**PURBANCHAL UNIVERSITY**  
**III SEMESTER FINAL EXAMINATION-2002**

**LEVEL:** B. E. (Civil/Computer/Electronics & Communication)

**SUBJECT:** BEG201HS, Mathematics-III

**TIME:** 3:00 hrs.

**Full Marks:** 80

**Pass Marks:** 32

*Candidates are required to give their answers in their own words as far as possible.*

*Figures in the margin indicate full marks.*

**GROUP-A**

**Attempt any THREE questions.**

**[3×10=30]**

Q. [1] (a) If A and B are square matrices of the same order prove that  $(AB)' = B' A'$ .

✓ (b) Evaluate: 
$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$$

Q. [2] (a) Express the matrix 
$$\begin{bmatrix} -1 & 7 & 1 \\ 2 & 3 & 4 \\ 3 & 0 & 5 \end{bmatrix}$$

as the sum of a symmetric and a skew-symmetric matrix.

(b) Prove that 
$$\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca).$$

*Contd. ...*



(2)

Q. [3] (a) Determine the rank of the matrix  $\begin{bmatrix} 2 & 6 & 0 \\ 6 & 20 & -6 \\ 0 & 6 & -18 \end{bmatrix}$ .

(b) Find the eigen value and eigen vectors of the matrix

$$\begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$$

Q. [4] (a) Verify Cayley-Hamilton theorem for the matrix  $\begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$ .

(b) Solve the system of equations using Gauss-elimination method:

$$x+y+z = 3$$

$$x+2y+3z = 4$$

$$x+4y+9z = 6$$

### GROUP-B

Attempt any TWO questions.

[2×10=20]

Q. [5] (a) Prove that the Laplace transform operator  $\alpha$  is a linear operator.

(b) Prove that  $\alpha \{ \cos \omega t \} = \frac{s}{s^2 + \omega^2}$ .

Q. [6] Determine the inverse Laplace transforms

(a)  $\alpha^{-1} \left( \frac{1}{s^2 - 4} \right)$

(b)  $\alpha^{-1} \left\{ \frac{1}{\sqrt{s+3}} \right\}$

Contd. ..

(3)

Q. [7] Solve  $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = 2e^{-t}$ ,  $y(0) = 2$ ,  $y'(0) = -1$

using Laplace transforms.

OR

Solve the initial value problem

$$y'' - 3y' + 2y = 4t + e^{3t}, \quad y(0) = 1, \quad y'(0) = -1$$

**GROUP-C**

Attempt any **THREE** questions.

[3×10=30]

Q. [8] (a) Prove that the line integral  $\int_C \vec{F} \cdot d\vec{r}$  is independent of the path joining any two points in the region if and only if  $\vec{F} = \nabla\Phi$  where  $\Phi$  is a scalar function.

(b) Evaluate  $\int_C \vec{F} \cdot d\vec{r}$  where  $\vec{F} = y^2\vec{i} + x^2\vec{j}$  and  $C$  is the straight line joining  $(0, 0)$  and  $(1, 1)$ .

Q. [9] (a) Show that the vector field

$$\vec{F} = (x^2 - yz)\vec{i} + (y^2 - zx)\vec{j} + (z^2 - xy)\vec{k}$$

is irrotational. Find a scalar  $\Phi$  such that  $\vec{F} = \nabla\Phi$ .

(b) Evaluate  $\iint_S \vec{F} \cdot \vec{n} \, ds$  where

$\vec{F} = (x - 2z)\vec{i} + (x + 3y + z)\vec{j} + (5x + y)\vec{k}$ ,  $S$  is the upper side of the triangle with vertices at  $(1, 0, 0)$ ,  $(0, 1, 0)$ , and  $(0, 0, 1)$ .

Contd. ...